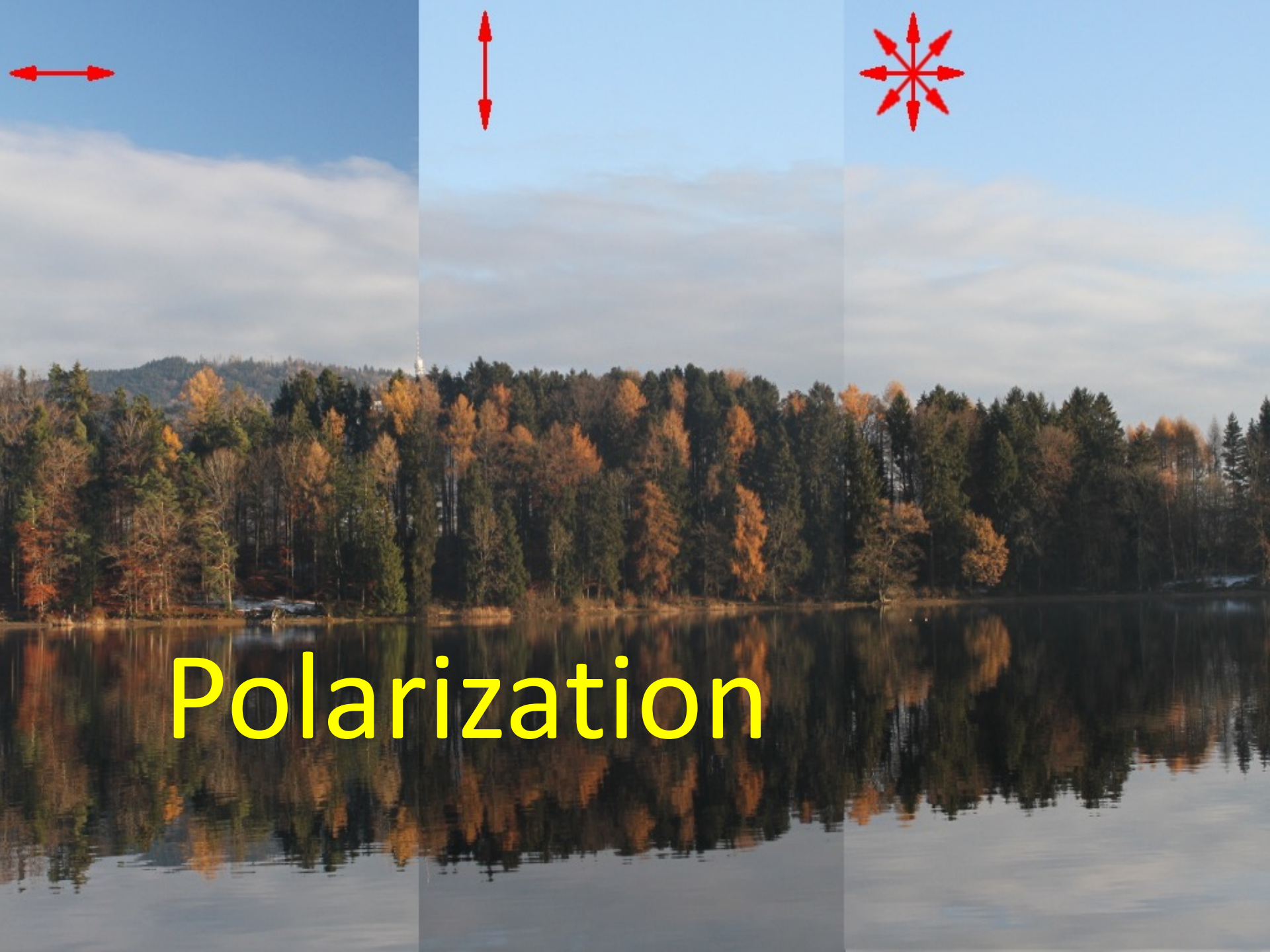
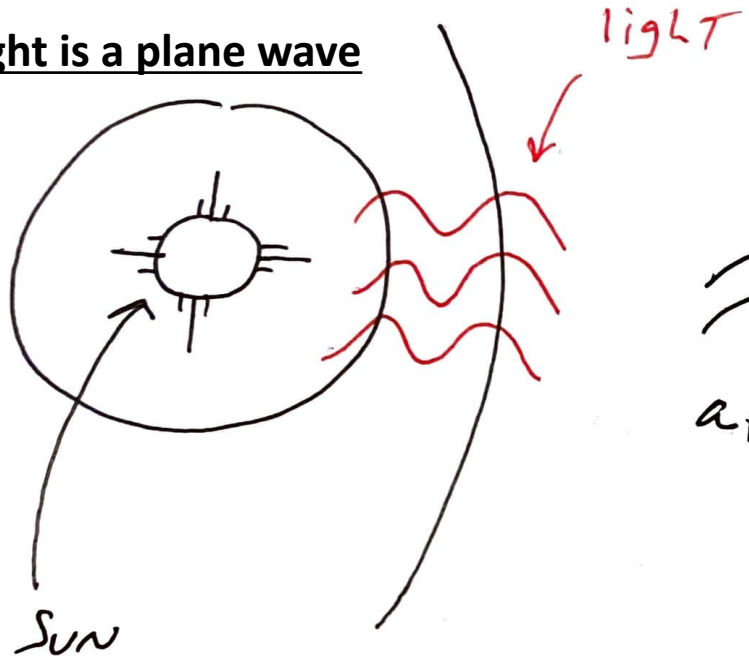
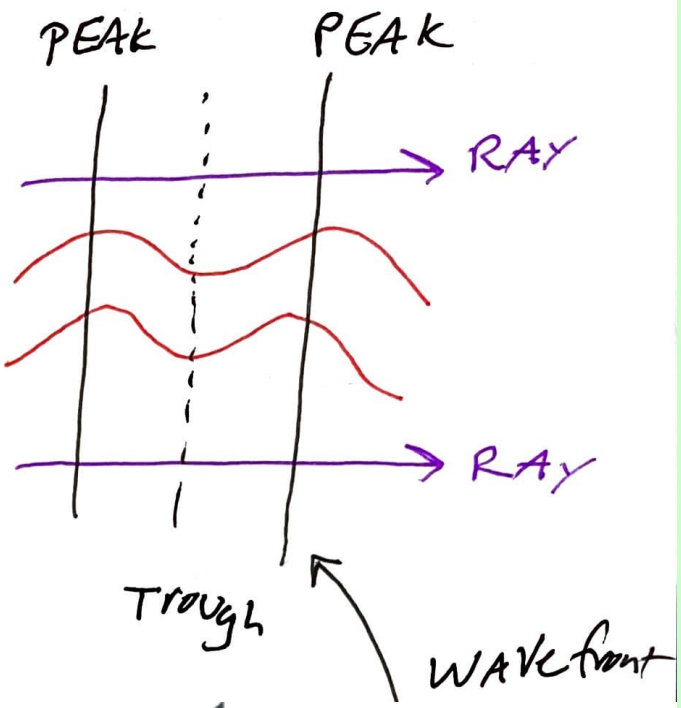

Polarization

Light is a plane wave

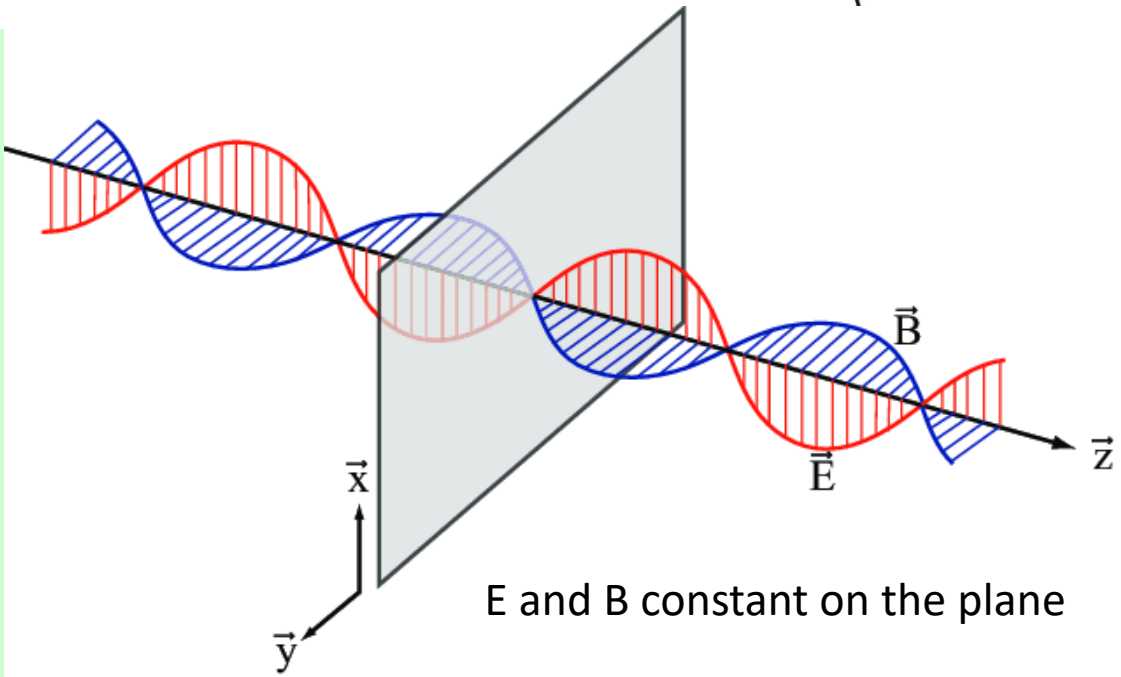


≈ approx.



$$E(x,t) = E_0 \sin(kx - \omega t + \phi_0)$$
$$B(x,t) = B_0 \sin(kx - \omega t + \phi_0)$$

E, B are in phase and normal



E and B constant on the plane

$E(x,t) = E_0 \sin(kx - \omega t + \phi_0)$ secretly encodes velocity..

$$v = \lambda f = \left(\frac{2\pi}{k}\right)\left(\frac{\omega}{2\pi}\right) = \boxed{\frac{\omega}{k}}$$

Light propagates at c only in a vacuum.
In all other media v<< c

Index of refraction:

(how many times slower in medium)

$$n = \boxed{\frac{c}{v}}$$

Material	n	Material	n
Vacuum	1.000	Ethyl alcohol	1.362
Air	1.000277	Glycerine	1.473
Water	4/3	Ice	1.31
Carbon disulfide	1.63	Polystyrene	1.59
Methylene iodide	1.74	Crown glass	1.50-1.62
Diamond	2.417	Flint glass	1.57-1.75

Ex. The magnetic field of a plane EM wave is given by:

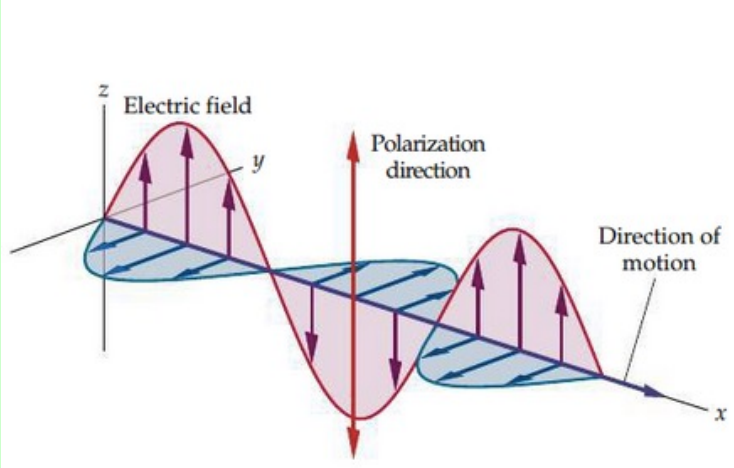
$$E(x,t) = 10\sin\left(2\times10^{7}x - 5\times10^{15}t\right)$$

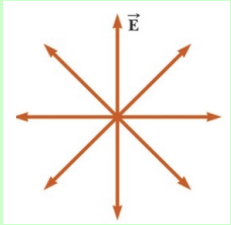
A. With which velocity does this EM wave propagate?
B. Determine the index of refraction? Is the EM wave in a vacuum?

Ex. How long does it take light to travel through an 3cm diamond?

Material	n	Material	n
Vacuum	1.000	Ethyl alcohol	1.362
Air	1.000277	Glycerine	1.473
Water	4/3	Ice	1.31
Carbon disulfide	1.63	Polystyrene	1.59
Methylene iodide	1.74	Crown glass	1.50-1.62
Diamond	2.417	Flint glass	1.57-1.75

Polarization:





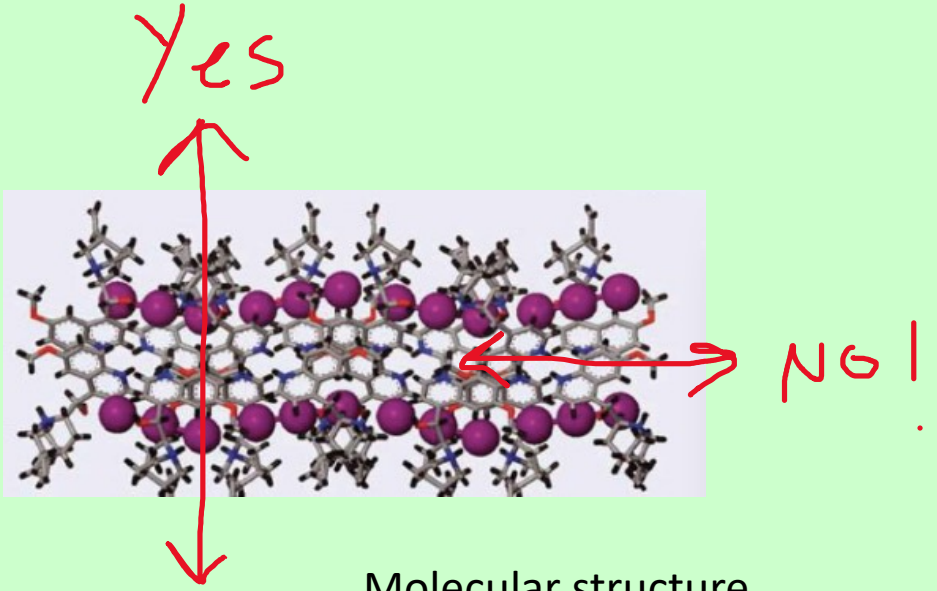
unpolarized



polarized

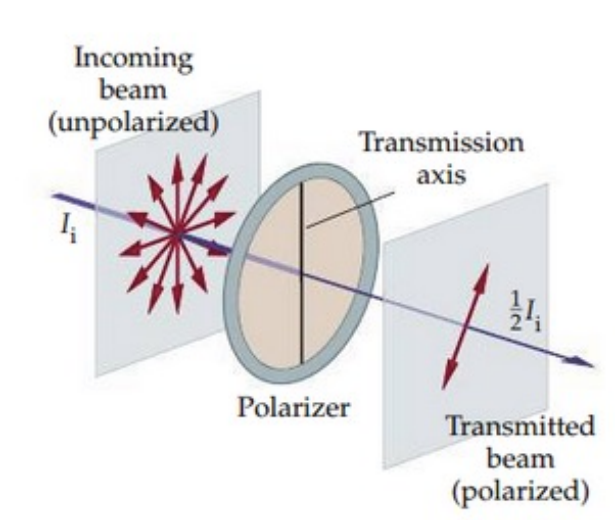


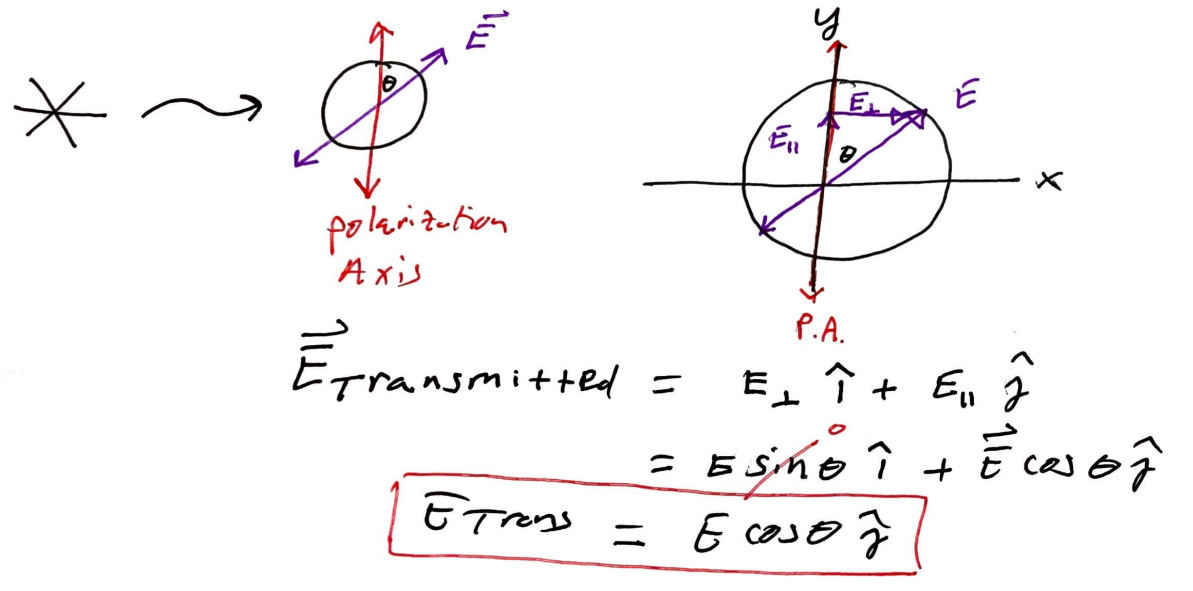
E.H. Land (1938) - Polaroid

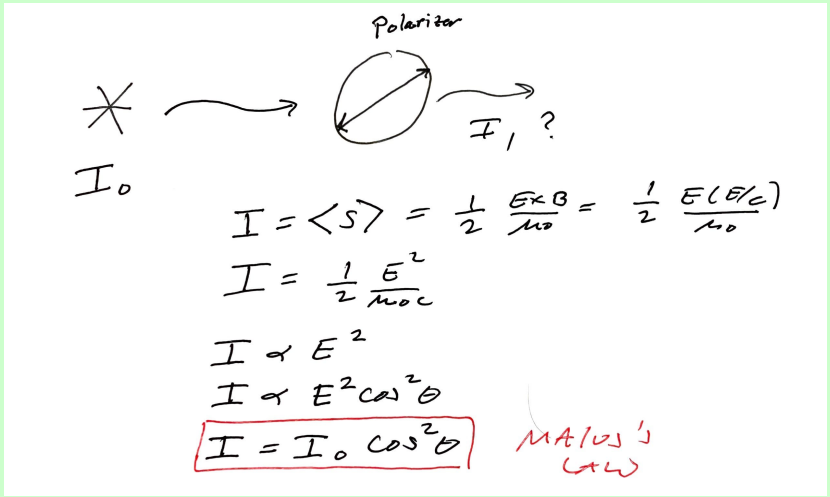


Molecular structure

Polarizers only transmit oscillations (E-fields) parallel to the polarization axis





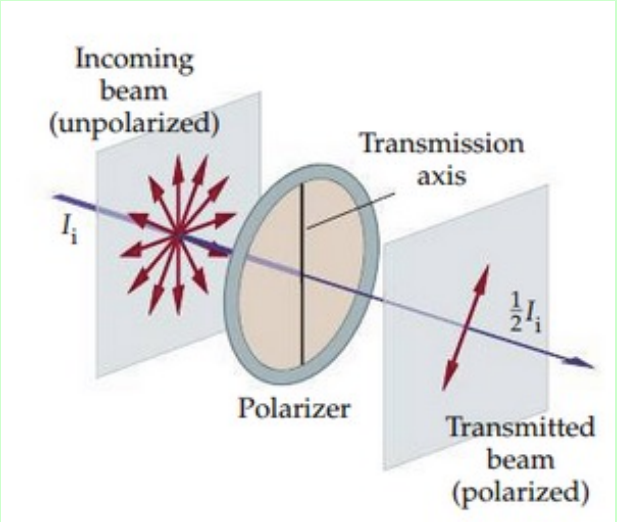


Malus's Law

$$I = I_0 \cos^2(\theta)$$

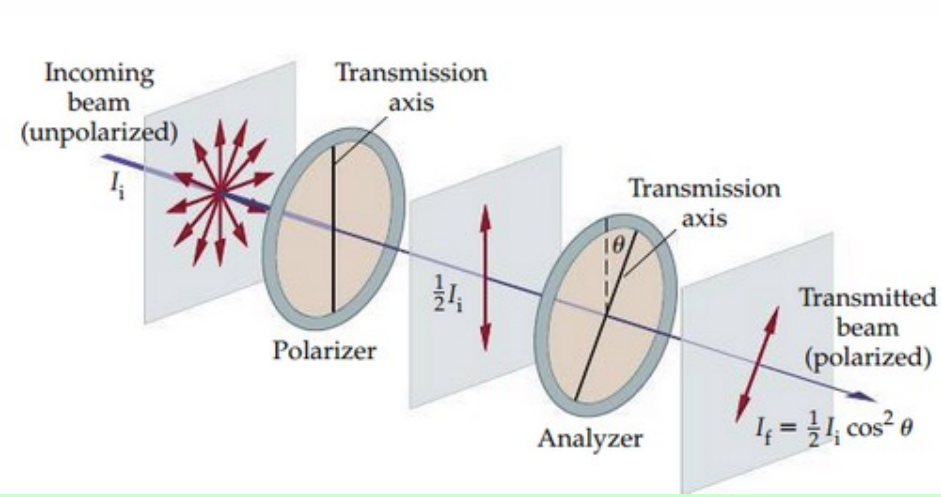
Ex. Use Malus's Law to show that the transmitted beam after going though a polarizer will be half as bright as the unpolarized beam

$$I = I_0 \cos^2(\theta)$$



c.f sec 1.3 page 3

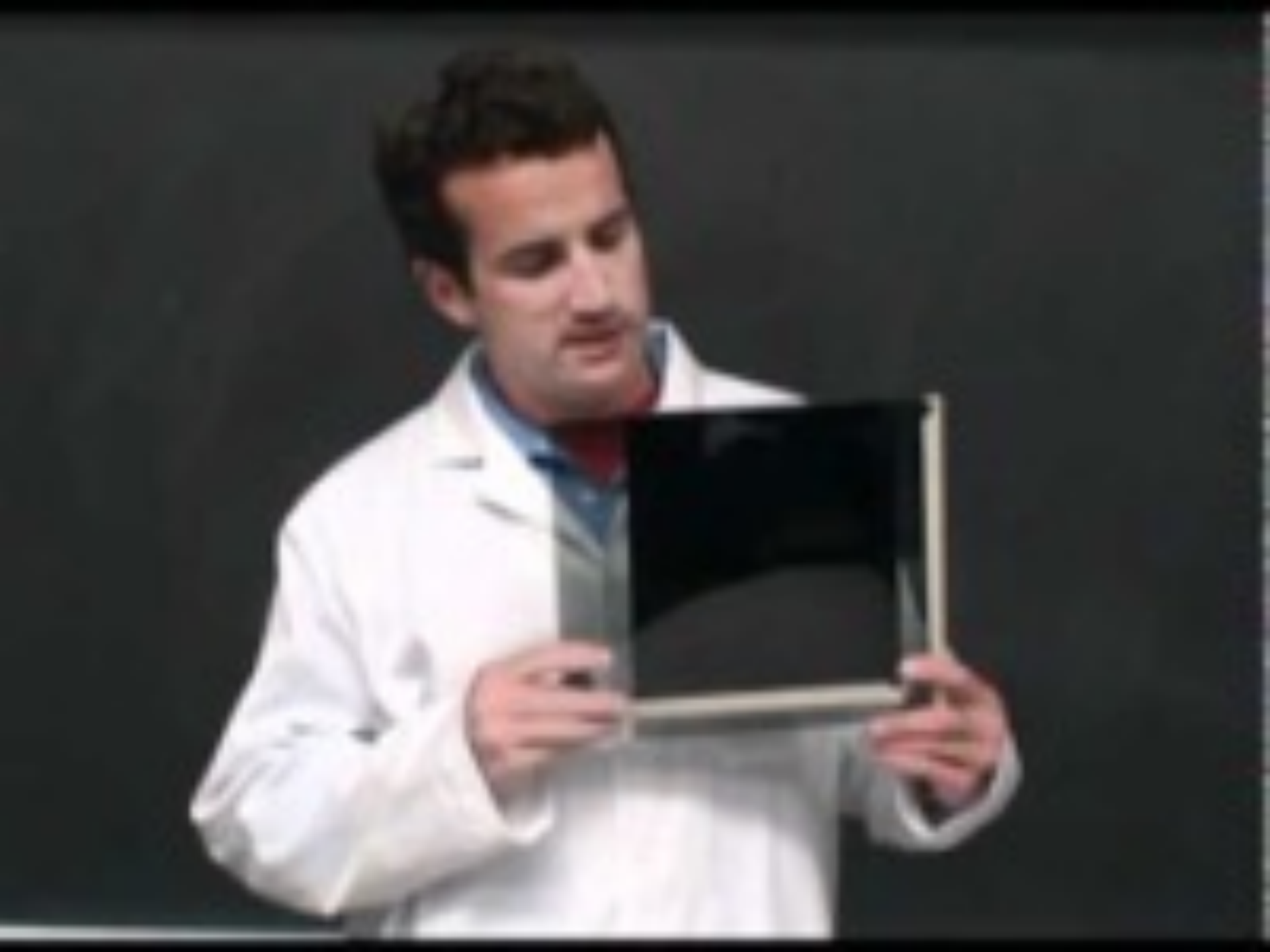
Ex. Unpolarized light is incident on a polarizer, and then a second polarizer whose polarization axis is 30 degrees rotated from the first



A. Determine the intensity of the outgoing beam relative to the first

Ex. Unpolarized light is incident on a set of two polarizers. The polarization axis of the second is rotated by 90 degrees from the first. What is the final intensity relative to the unpolarized beam?

Ex. Unpolarized light is incident on a set of three polarizers. The polarization axis of the second is rotated by 45 degrees from the first. The third polarizer is rotated by 90 degrees from the first. What is the final intensity relative to the unpolarized beam?



Cool sim

https://physics.bu.edu/~duffy/HTML5/polarized_light.html